

ER/EER Diagram Specification & Blueprint

Project: Mental Health Resource Accessibility Tracker

Course: Database Management Systems (CSE370)

1. Entities & Attributes

Core Entities & Keys

PATIENT

Primary Key (PK): patient_id

Simple Attributes: name, email, phone, date_of_birth, income_bracket, preferred_language, created_at, latitude, longitude

Composite Attribute: address (street, city, zip_code)

Derived Attribute: age (calculated from date_of_birth using a dashed oval)

PROVIDER (Superclass)

Primary Key (PK): provider_id

Simple Attributes: name, session_fee, max_capacity, rating_avg, latitude, longitude, accepts_insurance

Derived Attribute: utilization_rate (calculated from active bookings vs. max capacity)

LANGUAGE

Primary Key (PK): language_code

Simple Attributes: language_name

DISTRICT / REGION

Primary Key (PK): district_id

Simple Attributes: district_name, population, risk_index

SPECIALIZATION

Primary Key (PK): spec_id

Simple Attributes: spec_name (e.g., Anxiety, PTSD, Depression)

APPOINTMENT

Primary Key (PK): appointment_id

Simple Attributes: appointment_date, status (Completed, Cancelled, No-Show)

WAITLIST

Primary Key (PK): waitlist_id

Simple Attributes: request_date, crisis_score, priority_level, status

Derived Attribute: days_waiting (calculated from CURRENT_DATE - request_date)

REFERRAL

Primary Key (PK): referral_id

Simple Attributes: referral_date, status (Pending, Fulfilled, Expired), notes

RESOURCE_ACCESS_LOG

Primary Key (PK): log_id

Simple Attributes: resource_type, access_time

SYSTEM_ALERT

Primary Key (PK): alert_id

Simple Attributes: alert_type, created_at, status

2. EER Concepts: Specialization & Generalization

To meet higher-level EER diagram requirements for CSE370, use a Superclass / Subclass hierarchy for PROVIDER:

- Superclass: PROVIDER
Subclasses: THERAPIST, CLINIC, CRISIS_HOTLINE
- THERAPIST specific attributes: license_no, years_of_experience
CLINIC specific attributes: registration_no, total_beds
CRISIS_HOTLINE specific attributes: max_capacity, active_connections, status

Design Constraints

Disjointness Constraint: Disjoint (d) — A provider cannot be an individual Therapist, a Clinic, and a Crisis Hotline simultaneously.

Completeness Constraint: Total (t / double line) — Every Provider registered in the database must belong to one of these 3 specialized categories.

3. Relationships, Cardinalities & Participation

Relationship Name	Connecting Entities	Cardinality Ratio	Participation Constraint
Resides_In	PATIENT → DISTRICT	N : 1	Total on PATIENT (double line), Partial on DISTRICT
Located_In	PROVIDER → DISTRICT	N : 1	Total on PROVIDER (double line), Partial on DISTRICT

Offers	PROVIDER ↔ SPECIALIZATION	M : N	Partial on PROVIDER, Partial on SPECIALIZATION
Speaks	PROVIDER ↔ LANGUAGE	M : N	Partial on PROVIDER, Partial on LANGUAGE
Books	PATIENT → APPOINTMENT	1 : N	Partial on PATIENT, Total on APPOINTMENT
Conducts	PROVIDER → APPOINTMENT	1 : N	Partial on PROVIDER, Total on APPOINTMENT
Fulfills	REFERRAL → APPOINTMENT	1 : 1	Partial on REFERRAL, Partial on APPOINTMENT
Joins	PATIENT → WAITLIST	1 : N	Partial on PATIENT, Total on WAITLIST
Requires	SPECIALIZATION → WAITLIST	1 : N	Partial on SPECIALIZATION, Total on WAITLIST
Receives	PATIENT → REFERRAL	1 : N	Partial on PATIENT, Total on REFERRAL
Refers_From	PROVIDER (Source) → REFERRAL	1 : N	Partial on PROVIDER, Total on REFERRAL
Refers_To	PROVIDER (Target) → REFERRAL	1 : N	Partial on PROVIDER, Total on REFERRAL
Logs	PATIENT → RESOURCE_ACCESS_LOG	1 : N	Partial on PATIENT, Total on RESOURCE_ACCESS_LOG
Triggers	DISTRICT → SYSTEM_ALERT	1 : N	Partial on DISTRICT, Total on SYSTEM_ALERT